

Quiz 4

Imagine the chance experiment of drawing a poker hand (five cards) from a well shuffled standard deck of 52 cards. Let N be the random variable that counts the number of aces in your hand. The following table gives the distribution of N , with probability values rounded to four significant figures.

Value i of N	$P(N = i)$
0	0.6588
1	0.2995
2	0.03993
3	0.001736
4	0.00001847

1. Explain how to calculate $P(N = 2)$ exactly (as a fraction instead of approximate decimal).

A poker hand is a choice of 52 cards drawn from the 52 cards in the deck. There are $C_{52,5}$ possible poker hands. We assume that the deck is well shuffled so that all hands are equally likely. We can use the multiplication rule to count hands with exactly two aces. To make such a hand, first we choose 2 of the 4 aces in the deck and then 3 of the 48 non-ace cards. According to the multiplication rule, there are $C_{4,2}C_{48,3}$ possibilities for a hand with exactly 2 aces. The probability of drawing such a hand is thus the quotient $C_{4,2}C_{48,3} / C_{52,5}$, which works out to be approximately 0.03993.

2. Calculate the expected value of N using the distribution given in the table.

$$E[N] = (0)(0.6588) + (1)(0.2995) + (2)(0.03993) + (3)(0.001736) + (4)(0.00001847) \approx 0.3846.$$

We will learn later that it is easy to calculate the exact value of $E[N]$, which is $20/52$.

3. Calculate the variance and standard deviation of N using the distribution given in the table.

$$E[N^2] = (0^2)(0.6588) + (1^2)(0.2995) + (2^2)(0.03993) + (3^2)(0.001736) + (4^2)(0.00001847) \approx 0.4751$$

$$\text{Var}(N) = E[N^2] - E[N]^2 \approx 0.3272.$$

$$\sigma(N) = \sqrt{\text{Var}(N)} \approx 0.5720.$$

We will learn to compute $\text{Var}(N)$ (and variances of similar “hypergeometric” random variables) exactly and efficiently in Chapter 6.